

LESSON 109 (2.0) GROUND

Lesson Objective

During this lesson, the applicant will be introduced to the Piper Warrior systems. The applicant will gain a better understanding of the components and operations of Piper Warrior systems

Briefing

- Outline the objectives and completion standards of the lesson.
- Establish the student and instructor responsibilities.
- Review homework assigned in the previous lesson.
 - Quiz the student on the relevant sections of the Flight Operations Manual:
- Student questions.

References

- Pilot's Handbook of Aeronautical Knowledge. ([FAA-H-8083-25B](#))
 - Chapter 7: Aircraft Systems
- Florida Tech Aviation Systems presentation
 - Training drive → Power Points → Systems
- Piper Warrior Pilot Information Manual (PIM)

Instructor Notes

- It is **mandatory** that you and your student stop by the maintenance hangar and have a mechanic provide additional insight on any aircraft system(s) of your or your student's choice.
 - When closing out the lesson, make sure to include "visited maintenance hangar" or similar wording in the lesson comments and which mechanic(s) did your systems overview.
- To ensure your student has a strong understanding of the subject area, incorporate instructional videos and illustrations as needed. Utilize any spare system components available at the flight line, and take your student out to the aircraft to provide a practical, big-picture perspective.
- Depending on how new these concepts are to your student, you may need to adjust the depth of instruction, slow the pace, or incomplete the lesson.
- Use the PA-28-161 Systems PowerPoint on the training drive as your visual backbone, but supplement it with hands-on exposure whenever possible. Pointing to a diagram is far less effective than standing next to the actual aircraft and showing the student where the fuel strainer, oil filler, and static port are located.

Lesson content

- **Airframe**
 - Stressed skin structure, Semi-Monocoque design (uses a substructure for added strength).
 - Fuselage reinforced with vertical bulkheads and frames.
 - Skin out of aluminum alloy (lightweight metal).

- **Primary and secondary flight controls**
 - Primary controls: ailerons, elevator, rudder
 - Control inputs about the three axes of flight
 - Secondary controls: wing flaps (Piper Warrior — manually actuated)
 - Trim system operation and purpose
 - Control system linkage and cable operation

- **Powerplant and Propeller**
 - CL4HAND:
 - Carbureted
 - Lycoming O-360-D3G
 - Horizontally opposed
 - Air cooled
 - Normally aspirated
 - Direct Drive
 - Four-stroke engine operating cycle
 - Carbureted fuel-air mixture system
 - Fixed-pitch propeller characteristics
 - Engine cooling via airflow and baffles
 - Engine instruments (tachometer, oil pressure, oil temperature, CHT if equipped)

- **Fuel System**
 - Fuel tanks:
 - Constructed of aluminum.
 - Placed inboard leading edge of each wing.
 - Capacity:
 - 25 gallons per tank.
 - 24 gallons usable per tank.
 - Tabs indicate 17 gallons usable on the respective side.
 - Float device in each tank to measure quantity.
 - Float device is connected to an electrical sensor outside on back of tank.
 - Internal fuel strainer.
 - Fuel cap is vented.
 - Fuel vent tube under wing.

- Fuel Flow Path
 - Fuel flows from the selected tank through the selector valve.
 - It passes through a fuel strainer (gascolator) to remove contaminants.
 - Fuel then flows to the engine-driven fuel pump.
 - An electric auxiliary fuel pump is installed for backup and for certain operations (takeoff and landing as specified in the POH).
 - Fuel enters the carburetor where it is mixed with air before entering the engine cylinders.
- Minimum fuel grade:
 - 100 octane, we use 100 Low Lead.
 - Never use lower grade fuel. Always minimum or higher!
 - Always strain the drain and the fuel tanks before each flight to check for water or deposits.

● Oil System

- Wet sump lubrication system.
- The sump holds up to 8 quarts of SAE 50 W oil.
- 50 W = viscosity, or weight, of the oil (a liquid's resistance to flow).
- SAE = Society of Automotive Engineers.
- FIT recommendation:
 - 6 to 8 quarts.
- Manufacturer recommendation:
 - 2 to 8 quarts.
- Oil circulation
 - Oil is drawn from the sump through a suction screen by an engine-driven pump, forced through an oil filter (or screen)
 - Directed through an oil cooler (or bypassed), regulated by a pressure.
 - Distributed to lubricate internal engine parts before draining back to the sump.
- Purpose of lubrication, cooling, cleaning, sealing

● Hydraulic System

- Hydraulic system is used only for wheel braking.
- Toe brake pressure actuates master cylinders located behind each rudder pedal.
- Mechanical force is converted into hydraulic pressure using MIL-H-5606 fluid.
- Pressurized fluid travels through brake lines to calipers on each main landing gear wheel.
- Hydraulic pressure forces brake pads against the brake disc to slow the aircraft.
- Each main wheel operates on an independent hydraulic system, allowing differential braking.
- Parking brake system maintains hydraulic pressure to hold the aircraft stationary.

● **Electrical System**

- Battery
 - Used as a backup source of power in case the alternator fails.
 - 35 amperes.
 - 24 volts (some of the older warriors in the fleet have a 12 volts battery.)
- Alternator
 - 70 amperes.
 - 28 volts (some of the older warriors have a 14 volts alternator.)
 - Belt driven.
 - Located just behind the flywheel.
 - Primary source of electricity.
- Busbar
 - The busbar is the central electrical distribution point in the aircraft.
 - It receives electrical power from the battery and alternator.
 - Power is distributed from the main bus to aircraft systems, including lighting, pitot heat, fuel pump, and avionics.
- Circuit breaker
 - Thermal protection devices that trip when excessive current flows.
 - Can be reset once if condition is verified as non-hazardous.
- Fuses
 - Overcurrent protection devices that must be replaced if blown.
 - Not resettable in flight.
 - Less common in modern Warrior panels compared to circuit breakers.
- Voltage regulator
 - Regulates voltage between 28 volts and 31 volts.
 - 13-15 volts for the 14V alternator.
 - Overvoltage relay will open at 31 volts and remove the alternator from the circuit.
- Ammeter/Load meter
 - Displays load in amperes placed on the system.
 - With only the battery master switch on, the ammeter will show the amount of charging current demanded by the battery.
 - It never shows the discharge rate of the battery.
- Avionics
 - G5
 - Top screen used as PFD
 - Bottom screen used as HIS
 - Internal components
 - Accelerometers ▪ for the attitude indicator.
 - Magnetometer ▪ for the HSI.
 - Avidyne Integra
 - Air Data Computer (ADC):
 - Uses pitot-static system and OAT information for indicated

Enhanced Private Pilot Certification Course

- airspeed, true airspeed, altitude, and vertical speed.
 - Transducers sense air pressure and convert it to digital information.
- Attitude and Heading Reference System (AHRS):
 - Uses an accelerometer and magnetometer for attitude, rate of turn, and heading information.
 - Limitations are found in section 09 of the PIM.
- MFD
 - Data Acquisition Unit (DAU):
 - DAU receives information from pressure transducers and sensors throughout the engine.
 - MFD is able to show all of the engine information.
 - It has a temperature probe to monitor system temperatures.
 - Limitations are found in section 09 of the PIM.
- Autopilot: S-Tec 55x
 - Two-axis autopilot system providing roll and pitch control.
 - Controls heading and altitude hold functions.
 - Interfaces with navigation systems (VOR, GPS) for tracking courses.
 - Altitude hold mode maintains selected altitude.
 - Vertical speed mode (if installed) allows controlled climb/descent rate.
 - Electric trim integration reduces control pressures.
 - Autopilot disconnect available via control panel or yoke-mounted switch.
 - Autopilot is a supplemental aid; pilot maintains responsibility for aircraft control.
- **Ignition System**
 - Dual Magnetos
 - Impulse coupler
 - Pre-ignition vs Detonation
 - Engine starting process:
 - Rotating the key to the start position, activates the starter (battery powered).
 - Geared cog engages with the teeth of the flywheel.
 - Flywheel rotates, rotating the crankshaft.
 - This rotation drives the magnetos, providing a high voltage pulse to a distributor, which directs it to the spark plugs.
 - Engine starts, starter geared cog retracts from the flywheel.
- **Pitot-Static System**
 - Pitot tube supplies ram air pressure to the airspeed indicator.
 - Static port supplies static pressure to altimeter, airspeed indicator, and vertical speed indicator.
 - Airspeed indicator compares pitot pressure to static pressure.
 - Blocked pitot tube causes erroneous airspeed indications.

- Blocked static port affects altimeter, VSI, and airspeed readings.
 - Alternate static source may be installed to provide backup static pressure.
- **Vacuum System**
 - Powered by an engine-driven vacuum pump.
 - Provides suction to operate certain gyroscopic flight instruments.
 - Typically powers the attitude indicator and heading indicator in non-glass configurations.
 - Vacuum pressure gauge indicates system suction level.
 - Vacuum pump failure results in loss of affected gyro instruments.
- **Environmental controls**
 - Cabin heat system uses warmed air from a heat exchanger surrounding the exhaust system.
 - Cabin air vents provide outside air circulation.
 - Defroster directs warm air to windshield to improve visibility.
 - Carbon monoxide risk exists if exhaust system leaks occur.
 - Warrior is not pressurized or air-conditioned in standard configuration.
- **Anti-icing & Deicing systems**
 - Pitot heat prevents ice accumulation on pitot tube.
 - Carburetor heat prevents or removes carburetor ice.
 - Windshield defrost assists with interior fogging.
 - Piper Warrior is not certified for flight into known icing conditions.
 - Ice accumulation significantly degrades lift and increases drag.
- **Aircraft servicing**
 - Review PIM section 8: Aircraft Servicing & handling

Completion Standards

This lesson is complete when the applicant can:

1. Gain a basic understanding of the operation of Piper Warrior systems.